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1

Physiology and levels of function

COMP

You should be able to: Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained.

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2

Structure and function relationship

COMP

You should be able to: Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems.

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3 Homeostasis defined

COMP

You should be able to: Define homeostasis, regulated variable, and setpoint, and distinguish homeostasis from chemical equilibrium and from steady state.

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4 Feedback loop components

COMP

You should be able to: Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector, and response, and trace body temperature or blood glucose through every step.

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5 Negative and positive feedback

COMP

You should be able to: Distinguish negative from positive feedback by the direction of the response and give a physiological example of each, explaining why a positive feedback loop needs an outside event to end it.

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6 Feedforward control and acclimatization

COMP

You should be able to: Explain anticipatory feedforward control and acclimatization and identify which one is operating in a given scenario.

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7

Local and reflex control pathways

COMP

You should be able to: Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other.

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8

Mass balance

COMP

You should be able to: Apply the mass balance equation to a solute or to body water and calculate the intake, production, and output combination that holds the amount in the body constant.

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9 Units and unit conversion

COMP

You should be able to: Convert among the units used in physiology including molarity, osmolarity, milliequivalents, mmHg, liters per minute, and percent solutions.

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10 Graphing and data interpretation

COMP

You should be able to: Construct a labeled graph with the independent variable on the x axis, and read slope, direction, and trend from a physiological data set.

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11 Experimental design and controls

COMP

You should be able to: Identify the hypothesis, independent variable, dependent variable, and control condition in a physiology experiment and state what the control rules out.

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12 Measurement error and variability

COMP

You should be able to: Distinguish random from systematic error and explain why physiological measurements are repeated and averaged.

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